

COMP 3069

Computer Graphics



Lab 6:
Rendering Pipeline

Autumn 2024



The University of
Nottingham

Contents

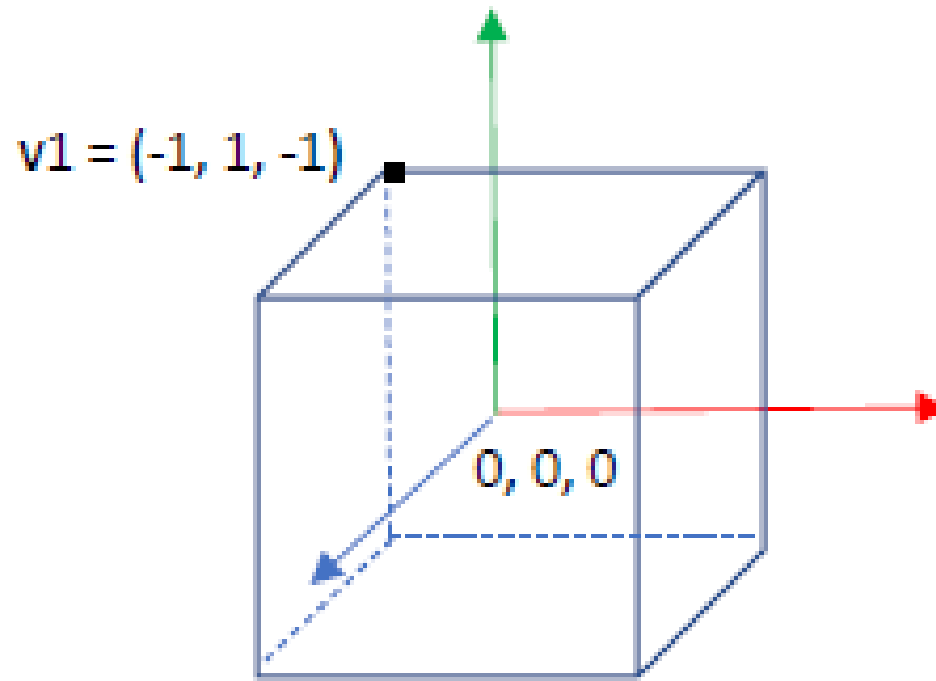
- ◆ Use a model, view and projection matrices
- ◆ Apply transformations to models and a camera
- ◆ Interpret different views of the same 3D scene (front, top, side and perspective views)
- ◆ Compose sequences of 3D transformations
- ◆ Diagnose possible incorrect transformations



Set-up of Lab6a

- ◆ Using the instructions in Lab 1, create Lab6a project
- ◆ Download `Lab6a.cpp` and copy the code to the `Lab6a.cpp` in the new project
- ◆ Copy the `window.h`, `util.h` and `shader.h`, and add them to the Header Files
- ◆ Download `mvp.vert` and `col.frag`, copy them, and add them to the Source Files

Model



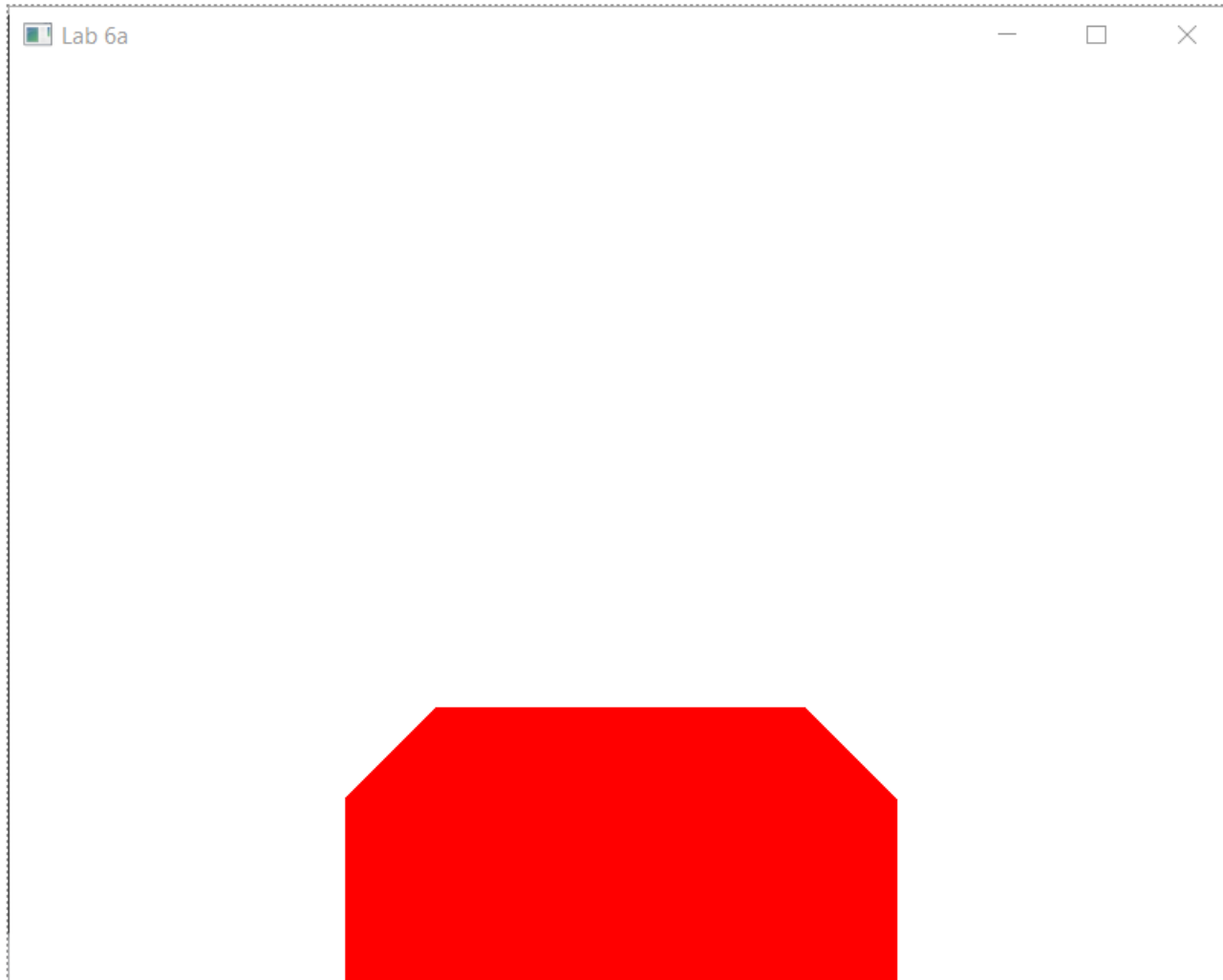
Code for Lab6a Default Display

```
17 float vertices[] =
18 {
19     //pos          //col
20     -1.0f, -1.0f, -1.0f, 1.0f, 0.0f, 0.0f,
21     1.0f, -1.0f, -1.0f, 1.0f, 0.0f, 0.0f,
22     1.0f, 1.0f, -1.0f, 1.0f, 0.0f, 0.0f,
23     1.0f, 1.0f, -1.0f, 1.0f, 0.0f, 0.0f,
24     -1.0f, 1.0f, -1.0f, 1.0f, 0.0f, 0.0f,
25     -1.0f, -1.0f, -1.0f, 1.0f, 0.0f, 0.0f,
26
27     -1.0f, -1.0f, 1.0f, 0.0f, 1.0f, 0.0f,
28     1.0f, -1.0f, 1.0f, 0.0f, 1.0f, 0.0f,
29     1.0f, 1.0f, 1.0f, 0.0f, 1.0f, 0.0f,
30     1.0f, 1.0f, 1.0f, 0.0f, 1.0f, 0.0f,
31     -1.0f, 1.0f, 1.0f, 0.0f, 1.0f, 0.0f,
32     -1.0f, -1.0f, 1.0f, 0.0f, 1.0f, 0.0f,
33
34     -1.0f, 1.0f, 1.0f, 0.0f, 0.0f, 1.0f,
35     -1.0f, 1.0f, -1.0f, 0.0f, 0.0f, 1.0f,
36     -1.0f, -1.0f, -1.0f, 0.0f, 0.0f, 1.0f,
37     -1.0f, -1.0f, -1.0f, 0.0f, 0.0f, 1.0f,
38     -1.0f, -1.0f, 1.0f, 0.0f, 0.0f, 1.0f,
39     -1.0f, 1.0f, 1.0f, 0.0f, 0.0f, 1.0f,
40
41     1.0f, 1.0f, 1.0f, 1.0f, 1.0f, 0.0f,
42     1.0f, 1.0f, -1.0f, 1.0f, 1.0f, 0.0f,
43     1.0f, -1.0f, -1.0f, 1.0f, 1.0f, 0.0f,
44     1.0f, -1.0f, -1.0f, 1.0f, 1.0f, 0.0f,
45     1.0f, -1.0f, 1.0f, 1.0f, 1.0f, 0.0f,
46     1.0f, 1.0f, 1.0f, 1.0f, 1.0f, 0.0f,
47
48     -1.0f, -1.0f, -1.0f, 1.0f, 0.0f, 1.0f,
49     1.0f, -1.0f, -1.0f, 1.0f, 0.0f, 1.0f,
50     1.0f, -1.0f, 1.0f, 1.0f, 0.0f, 1.0f,
51     1.0f, -1.0f, 1.0f, 1.0f, 0.0f, 1.0f,
52     -1.0f, -1.0f, 1.0f, 1.0f, 0.0f, 1.0f,
53     -1.0f, -1.0f, -1.0f, 1.0f, 0.0f, 1.0f,
54
55     -1.0f, 1.0f, -1.0f, 0.0f, 1.0f, 1.0f,
56     1.0f, 1.0f, -1.0f, 0.0f, 1.0f, 1.0f,
57     1.0f, 1.0f, 1.0f, 0.0f, 1.0f, 1.0f,
58     1.0f, 1.0f, 1.0f, 0.0f, 1.0f, 1.0f,
59     -1.0f, 1.0f, 1.0f, 0.0f, 1.0f, 1.0f,
60     -1.0f, 1.0f, -1.0f, 0.0f, 1.0f, 1.0f,
61 };
```

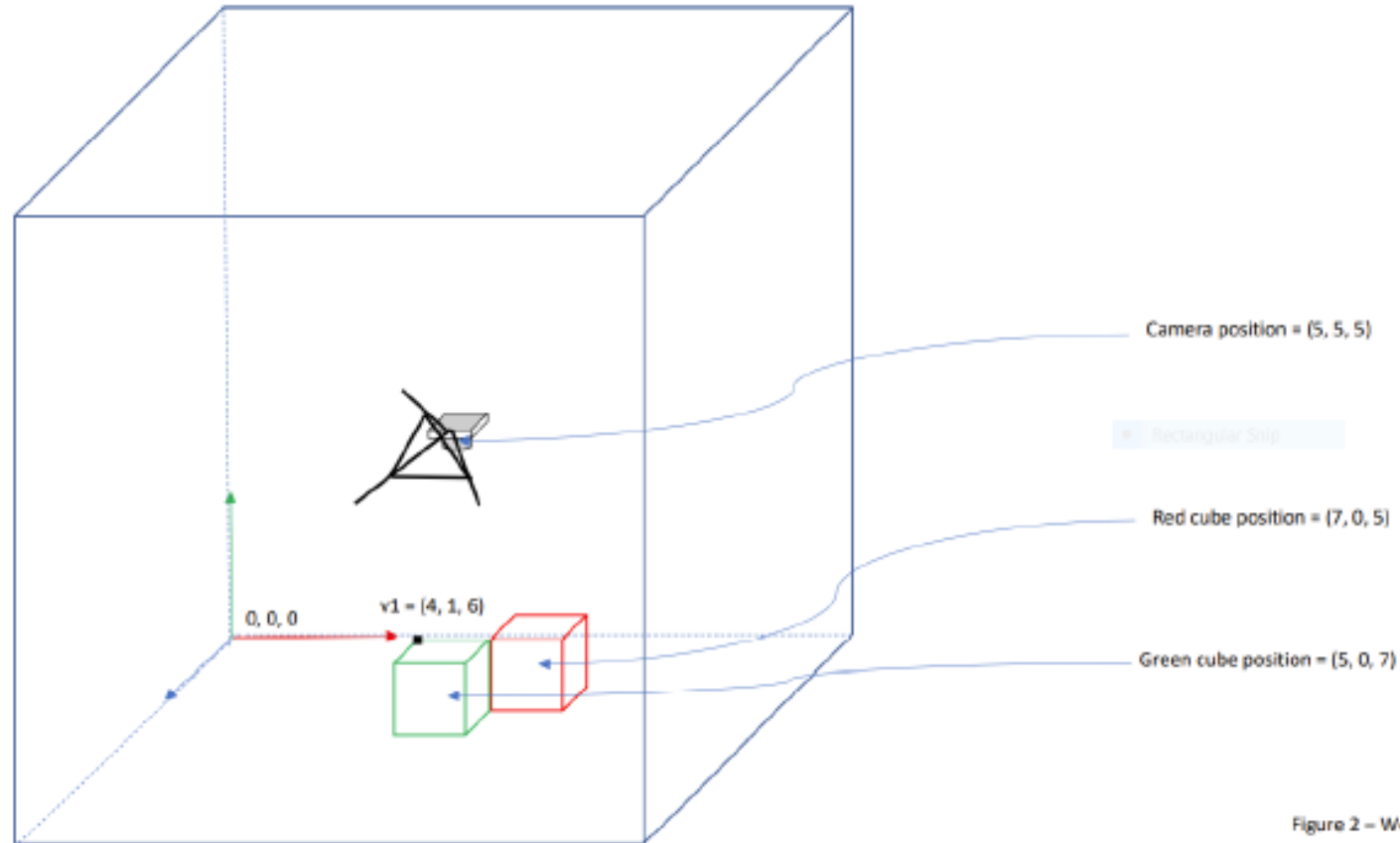
```
123 glm::mat4 view = glm::mat4(1.f);
124 view = glm::translate(view, glm::vec3(0.0f, 2.f, 5.0f));
125 int vloc = glGetUniformLocation(shaderProgram, "view");
126 glUniformMatrix4fv(vloc, 1, GL_FALSE, glm::value_ptr(view));
127
128 glm::mat4 projection = glm::mat4(1.f);
129 projection = glm::perspective(glm::radians(45.f), (float)800 / (float)600, 1.f, 10.f);
130 int ploc = glGetUniformLocation(shaderProgram, "projection");
131 glUniformMatrix4fv(ploc, 1, GL_FALSE, glm::value_ptr(projection));
132
133 int cloc = glGetUniformLocation(shaderProgram, "uCol");
134 glUniform3f(cloc, 1.f, 0.f, 0.f);
135
136 glm::mat4 model = glm::mat4(1.f);
137 int mloc = glGetUniformLocation(shaderProgram, "model");
138 glUniformMatrix4fv(mloc, 1, GL_FALSE, glm::value_ptr(model));
139 glBindVertexArray(VAO);
140 glDrawArrays(GL_TRIANGLES, 0, 36);
141
142 cloc = glGetUniformLocation(shaderProgram, "uCol");
143 glUniform3f(cloc, 0.f, 1.f, 0.f);
144
145 model = glm::mat4(1.f);
146 mloc = glGetUniformLocation(shaderProgram, "model");
147 glUniformMatrix4fv(mloc, 1, GL_FALSE, glm::value_ptr(model));
148 glBindVertexArray(VAO);
149 glDrawArrays(GL_TRIANGLES, 0, 36);
```



Lab6a Default Display



Task 1: Models and Camera

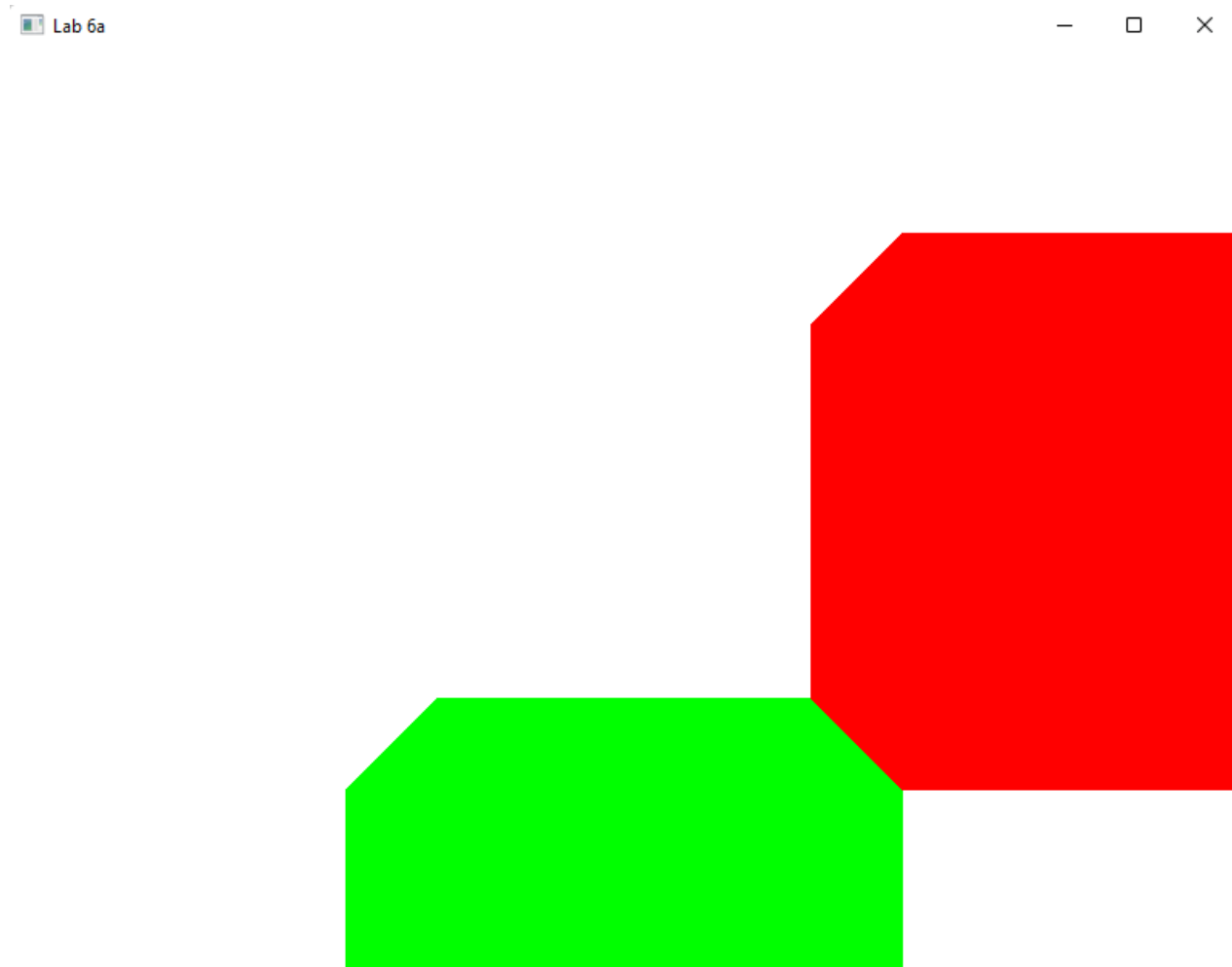


Task 1: Set up Models and View

```
123 glm::mat4 view = glm::mat4(1.f);
124 //view = glm::translate(view, -glm::vec3(0.0f, 2.f, 5.0f));
125 view = glm::rotate(view, (float)glm::radians(90.f), glm::vec3(1.f, 0.f, 0.f));
126 view = glm::translate(view, -glm::vec3(5.0f, 5.f, 5.0f));
127 int vloc = glGetUniformLocation(shaderProgram, "view");
128 glUniformMatrix4fv(vloc, 1, GL_FALSE, glm::value_ptr(view));
129
137
138 glm::mat4 model = glm::mat4(1.f);
139 model = glm::translate(model, glm::vec3(7.0f, 0.f, 5.0f));
140 int mloc = glGetUniformLocation(shaderProgram, "model");
141 glUniformMatrix4fv(mloc, 1, GL_FALSE, glm::value_ptr(model));
142 glBindVertexArray(VAO);
143 glDrawArrays(GL_TRIANGLES, 0, 36);
144
145 cloc = glGetUniformLocation(shaderProgram, "uCol");
146 glUniform3f(cloc, 0.f, 1.f, 0.f);
147
148 model = glm::mat4(1.f);
149 model = glm::translate(model, glm::vec3(5.0f, 0.f, 7.0f));
150 mloc = glGetUniformLocation(shaderProgram, "model");
151 glUniformMatrix4fv(mloc, 1, GL_FALSE, glm::value_ptr(model));
152 glBindVertexArray(VAO);
153 glDrawArrays(GL_TRIANGLES, 0, 36);
```



Task 1 Outcomes



Set-up of Lab6b

- ◆ Using the instruction in Lab 1, create Lab6b project
- ◆ Download `Lab6b.cpp` and copy the code to the `Lab6b.cpp` in the new project
- ◆ Copy the `window.h`, `util.h` and `shader.h`, and add them to the Header Files
- ◆ Download `mvp1.vert` and `coll.frag`, copy them, and add them to the Source Files

Code for Lab6b Default Display

```

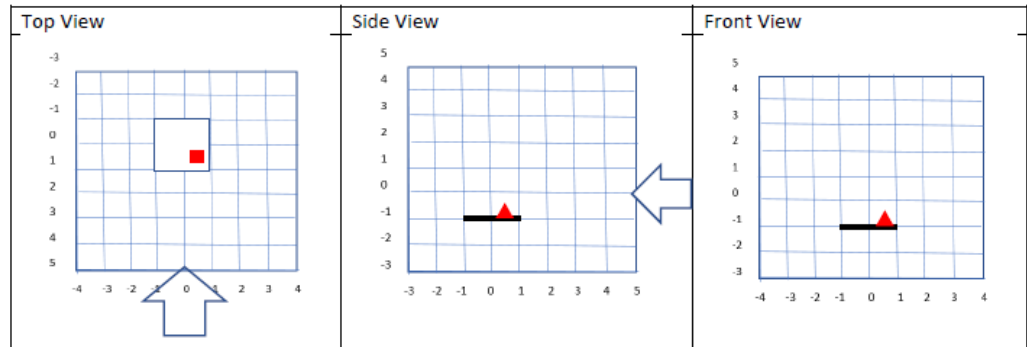
63 float vertices2[] =
64 {
65     //BASE
66     //pos                //col
67     -1.0f, -1.0f, 1.0f,    1.f, 1.0f, 1.0f,
68     -1.0f, -1.0f, -1.0f,  1.f, 1.0f, 1.0f,
69     1.0f, -1.0f, -1.0f,   1.f, 1.0f, 1.0f,
70     -1.0f, -1.0f, 1.0f,   1.f, 1.0f, 1.0f,
71     1.0f, -1.0f, -1.0f,   1.f, 1.0f, 1.0f,
72     1.0f, -1.0f, 1.0f,    1.f, 1.0f, 1.0f,
73
74     //TL
75     //pos                //col
76     .25f, -1.0f, .25f,    1.f, 0.0f, 0.0f,
77     0.5f, -.75f, 0.5f,    1.0f, 0.0f, 0.0f,
78     .25f, -1.0f, .75f,    1.f, 0.0f, 0.0f,
79
80     //TB
81     //pos                //col
82     .75f, -1.0f, .25f,    1.f, 0.0f, 0.0f,
83     0.5f, -.75f, 0.5f,    1.0f, 0.0f, 0.0f,
84     .25f, -1.0f, .75f,    1.f, 0.0f, 0.0f,
85
86     //TR
87     //pos                //col
88     .75f, -1.0f, .75f,    1.0f, 0.0f, 0.0f,
89     0.5f, -.75f, 0.5f,    1.0f, 0.0f, 0.0f,
90     .75f, -1.0f, .25f,    1.0f, 0.0f, 0.0f,
91
92     //TF
93     //pos                //col
94     .75f, -1.0f, .75f,    1.f, 0.0f, 0.0f,
95     0.5f, -.75f, 0.5f,    1.0f, 0.0f, 0.0f,
96     .25f, -1.0f, .75f,    1.f, 0.0f, 0.0f,
97 };

```

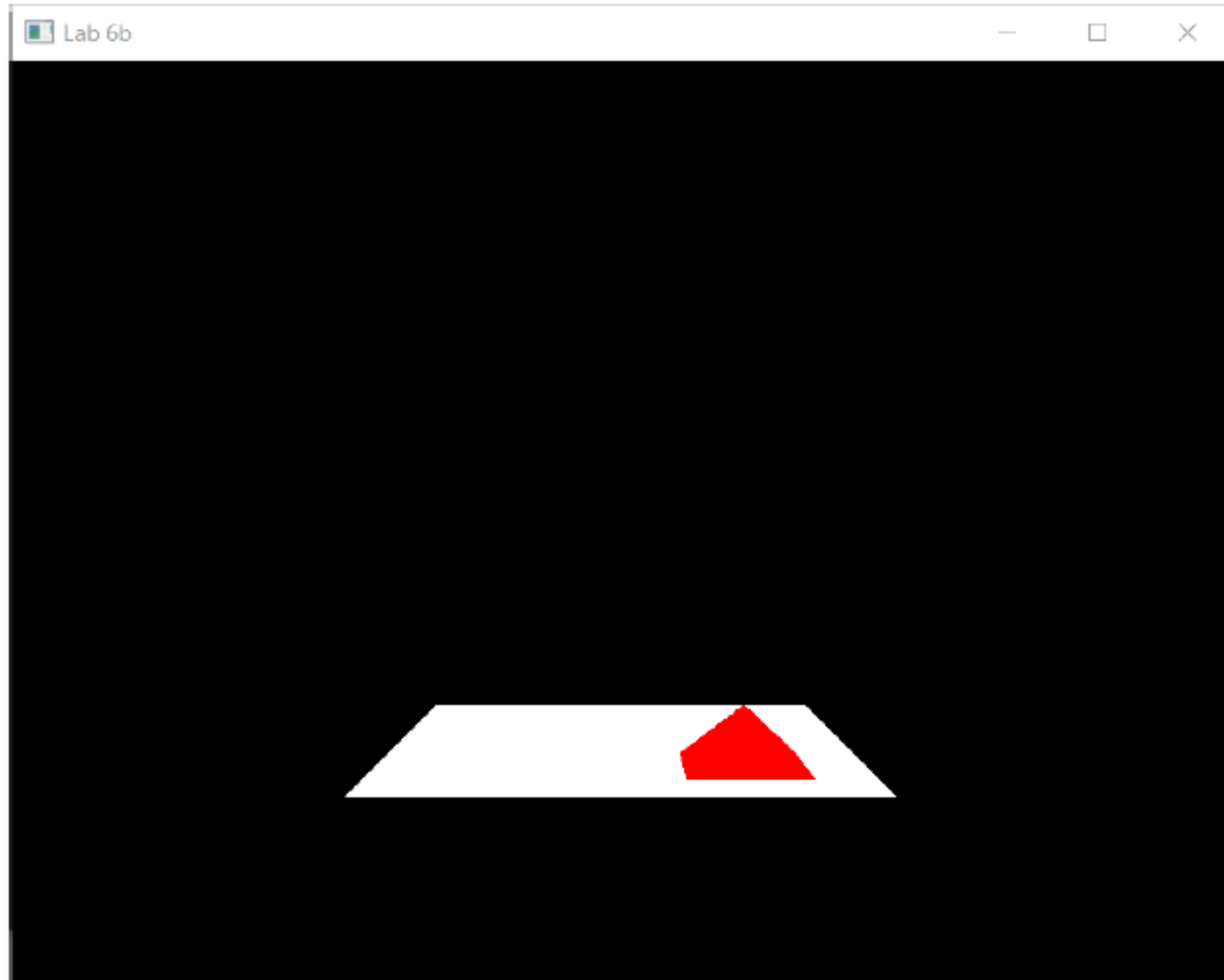
```

155
156 glm::mat4 model = glm::mat4(1.f);
157 model = glm::translate(model, glm::vec3(0.f, 0.f, 0.0f));
158 int mloc = glGetUniformLocation(shaderProgram, "model");
159 glUniformMatrix4fv(mloc, 1, GL_FALSE, glm::value_ptr(model));
160
161 glm::mat4 view = glm::mat4(1.f);
162 view = glm::translate(view, -glm::vec3(0.0f, 0.0f, 5.0f));
163 int vloc = glGetUniformLocation(shaderProgram, "view");
164 glUniformMatrix4fv(vloc, 1, GL_FALSE, glm::value_ptr(view));
165
166 glm::mat4 projection = glm::mat4(1.f);
167 projection = glm::perspective(glm::radians(45.f), (float)800 / (float)600, .1f, 10.f);
168 int ploc = glGetUniformLocation(shaderProgram, "projection");
169 glUniformMatrix4fv(ploc, 1, GL_FALSE, glm::value_ptr(projection));

```

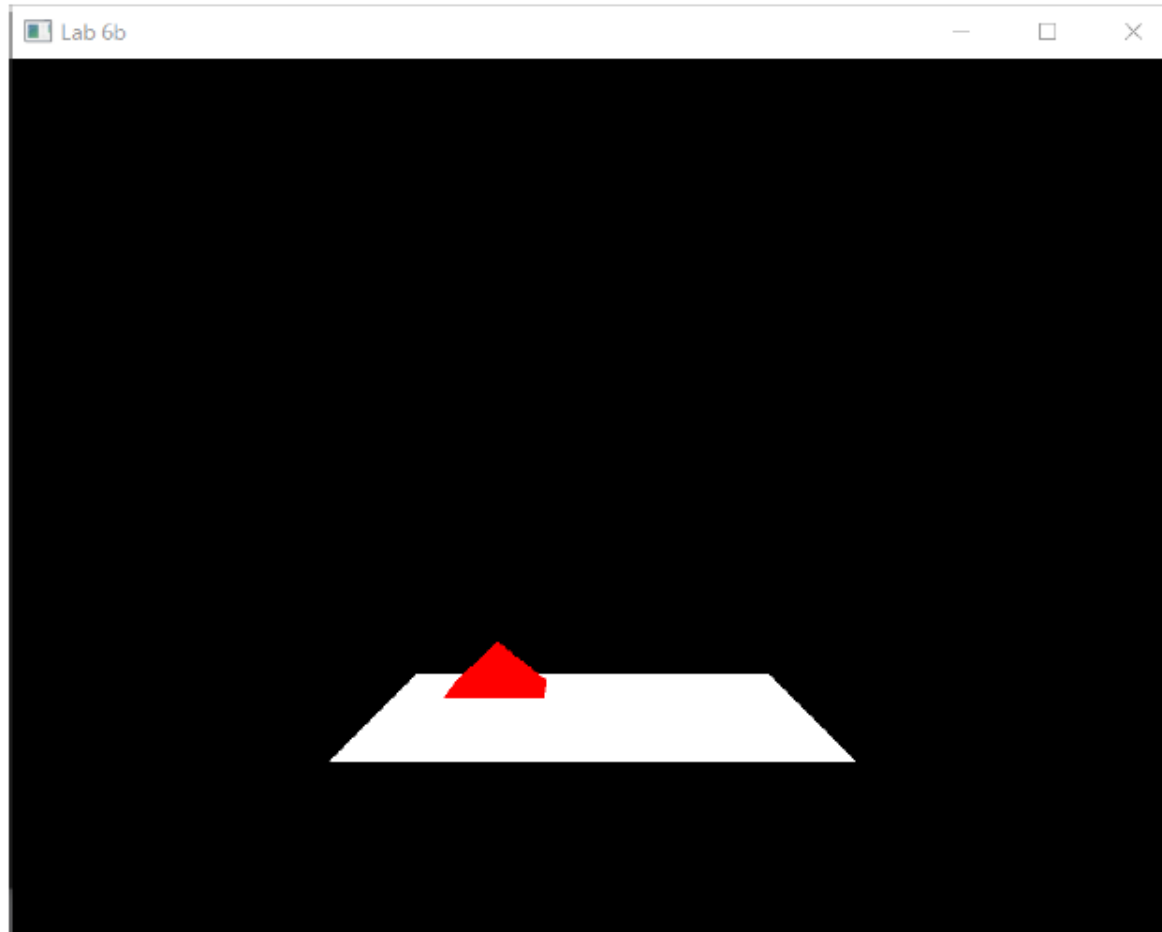


Lab6bDefault Display



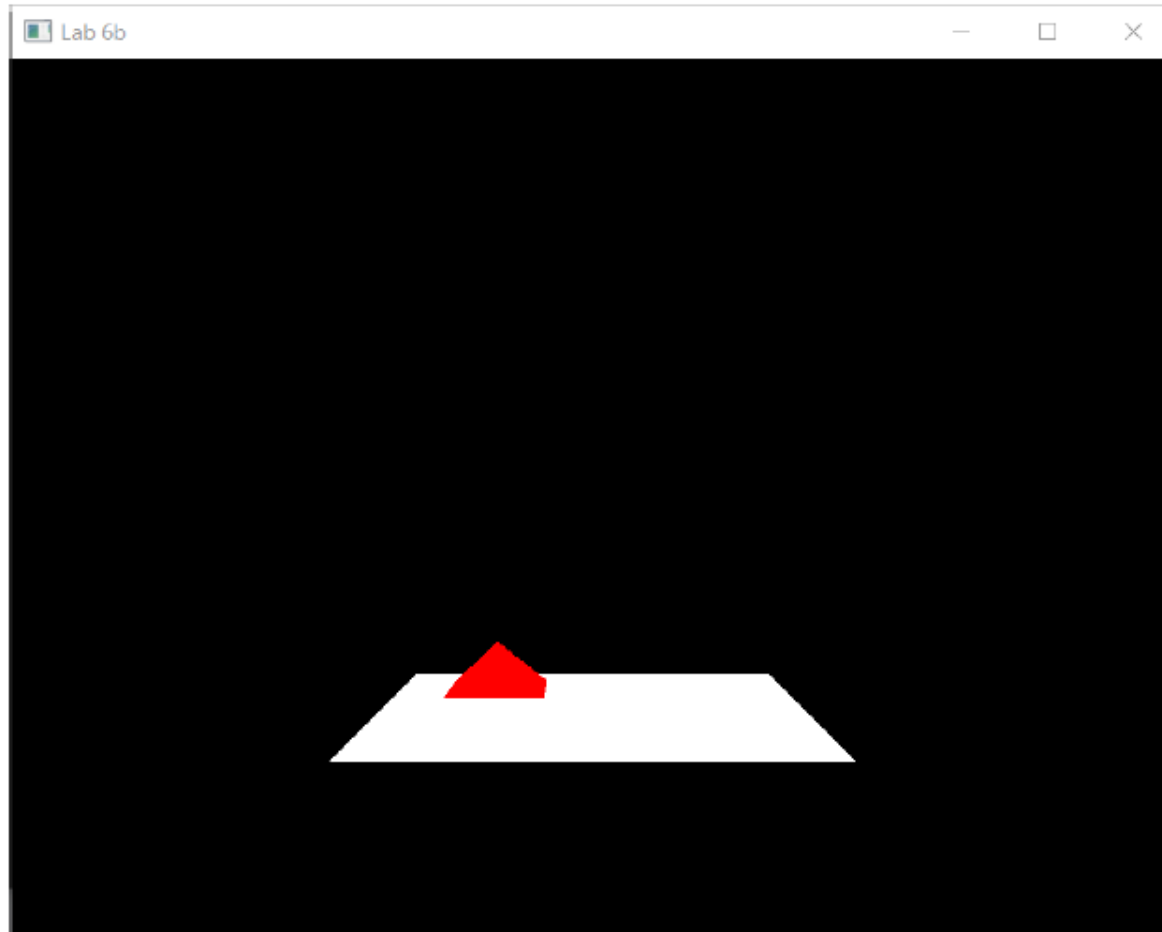
Task 2: Rotation by 180 Degrees

```
glm::mat4 model = glm::mat4(1.f);  
model = glm::rotate(model, (float)glm::radians(180.f), glm::vec3(0.f, 1.f, 0.f));
```



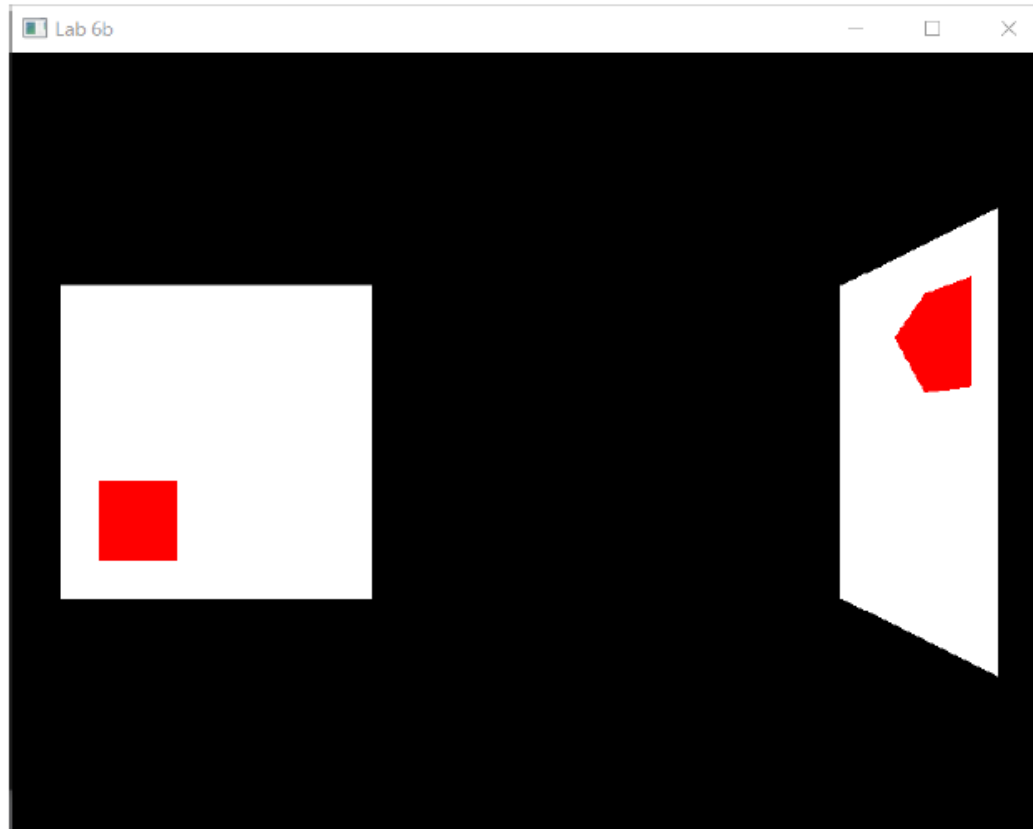
Task 2: Rotation by Scaling

```
glm::mat4 model = glm::mat4(1.f);  
model = glm::scale(model, glm::vec3(-1.f, 1.f, -1.f));
```



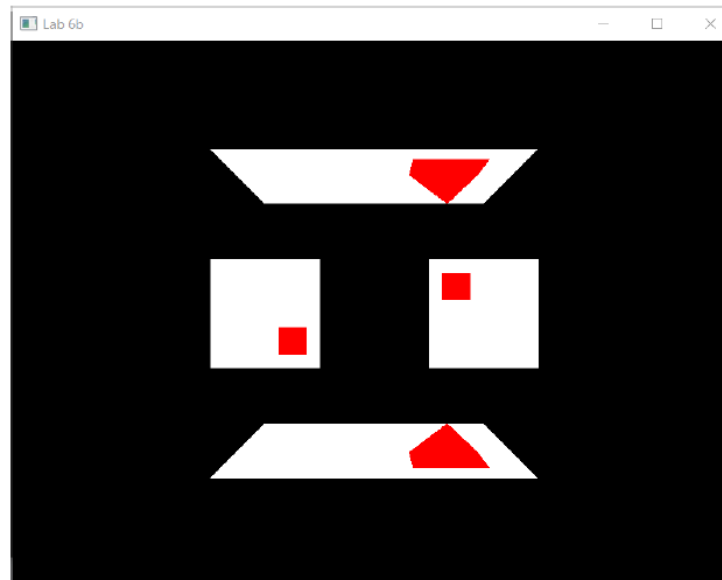
Task 4: 2 Models with Transformations

```
model = glm::translate(model, glm::vec3(-2.0f, 0.0f, .0f));  
model = glm::scale(model, glm::vec3(-1.0f, 1.0f, 1.0f));  
model = glm::rotate(model, (float)glm::radians(90.f), glm::vec3(1.0f, 0.0f, .0f));  
  
model = glm::mat4(1.f);  
model = glm::translate(model, glm::vec3(1.0f, .0f, .0f));  
model = glm::rotate(model, (float)glm::radians(90.f), glm::vec3(.0f, 0.0f, 1.0f));
```

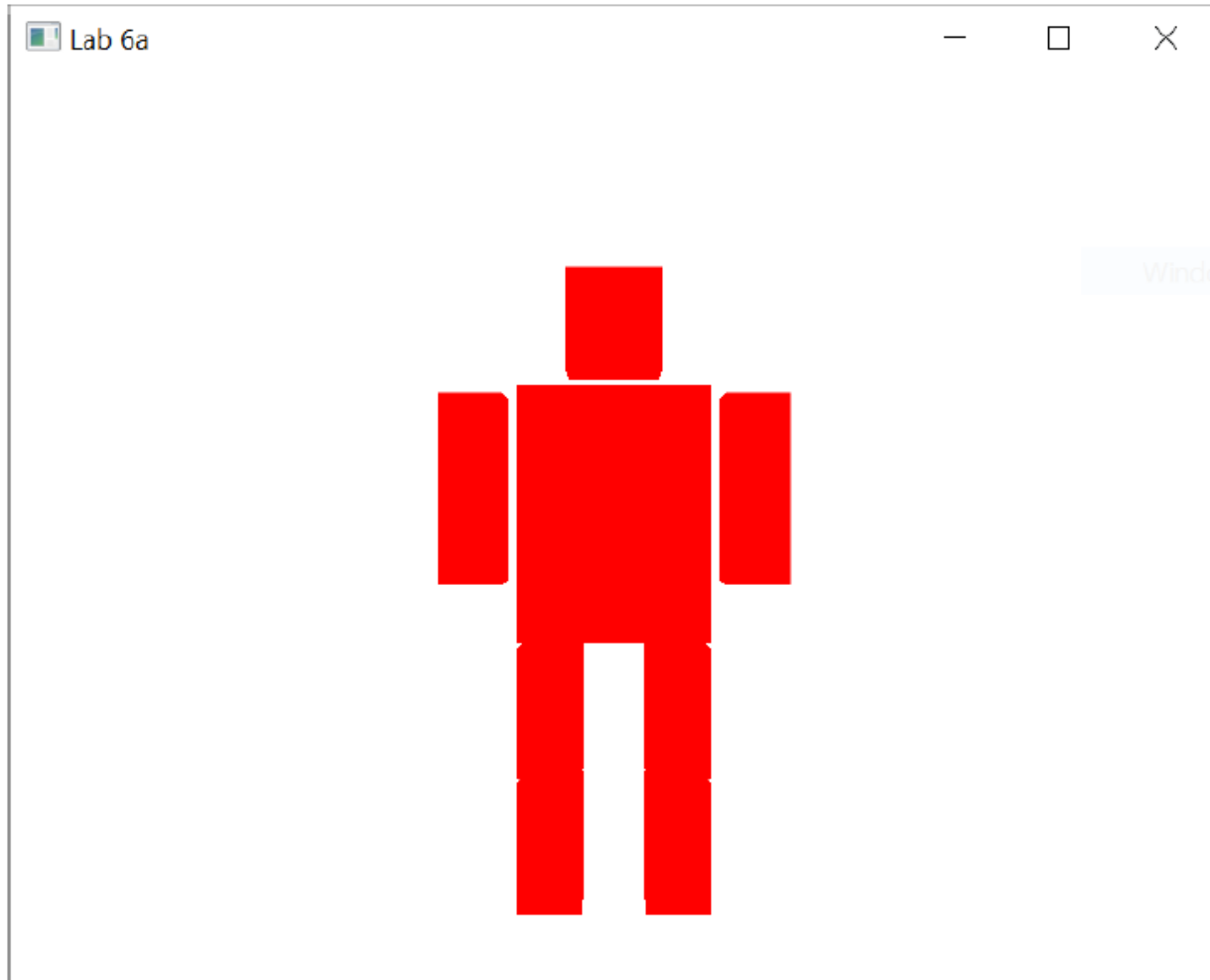


Task 5: 4 Models with Transformations

```
glm::mat4 model = glm::mat4(1.f);  
model = glm::scale(model, glm::vec3(1.f, -1.f, 1.f));  
  
model = glm::mat4(1.f);  
model = glm::translate(model, glm::vec3(-1.0f, .0f, 0.0f));  
model = glm::rotate(model, (float)glm::radians(90.f), glm::vec3(1.f, 0.f, 0.f));  
model = glm::scale(model, glm::vec3(.5f, 1.f, .5f));  
  
model = glm::mat4(1.f);  
model = glm::translate(model, glm::vec3(1.0f, .0f, 0.0f));  
model = glm::rotate(model, (float)glm::radians(180.f), glm::vec3(0.f, 0.f, 1.f));  
model = glm::rotate(model, (float)glm::radians(90.f), glm::vec3(1.f, 0.f, 0.f));  
model = glm::scale(model, glm::vec3(.5f, 1.f, .5f));  
  
model = glm::mat4(1.f);
```



Task 6: Option



References

- ◆ *COMP3069 Lab6 notes*
- ◆ *Interactive Computer Graphics: A Top-Down Approach with Shader-Based OpenGL, Chapter 6*